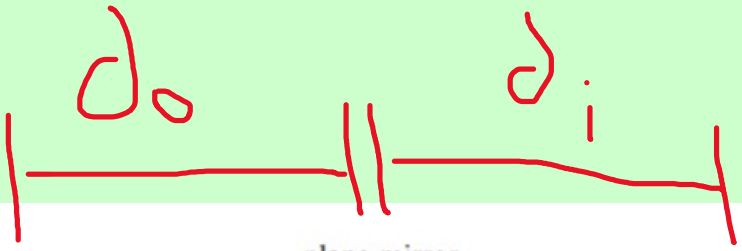
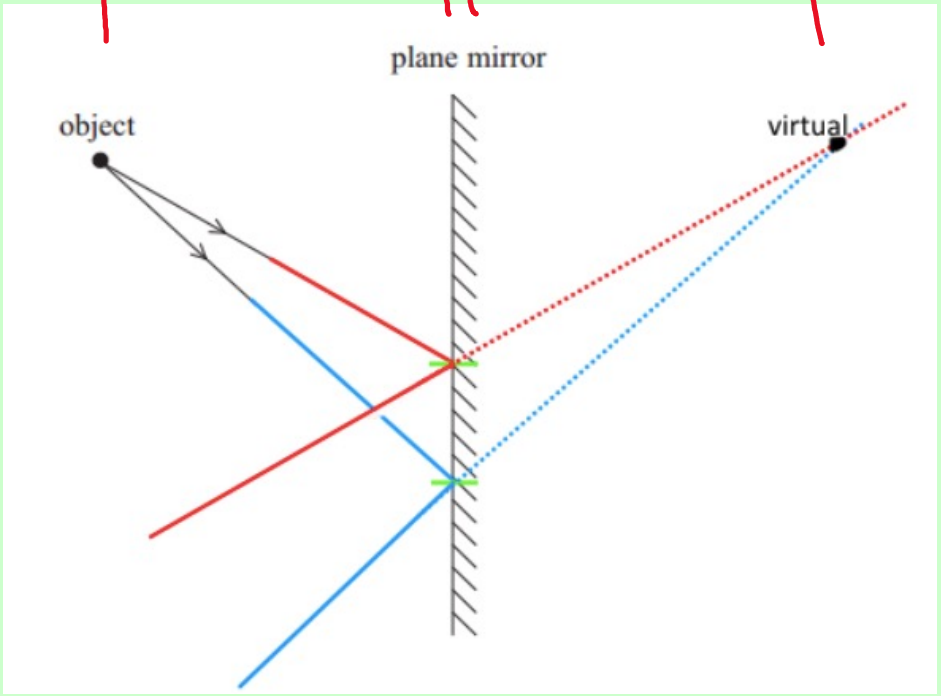

Mirrors

Recall mirrors…

Mirror: Brain assumes straight lines





Where rays converge, an **image** is formed

Real: places rays go
Virtual: Places they appear to go

d_0 Is the object distance

d_i Is the image distance

(for a flat mirror) $|d_0| = |d_i|$

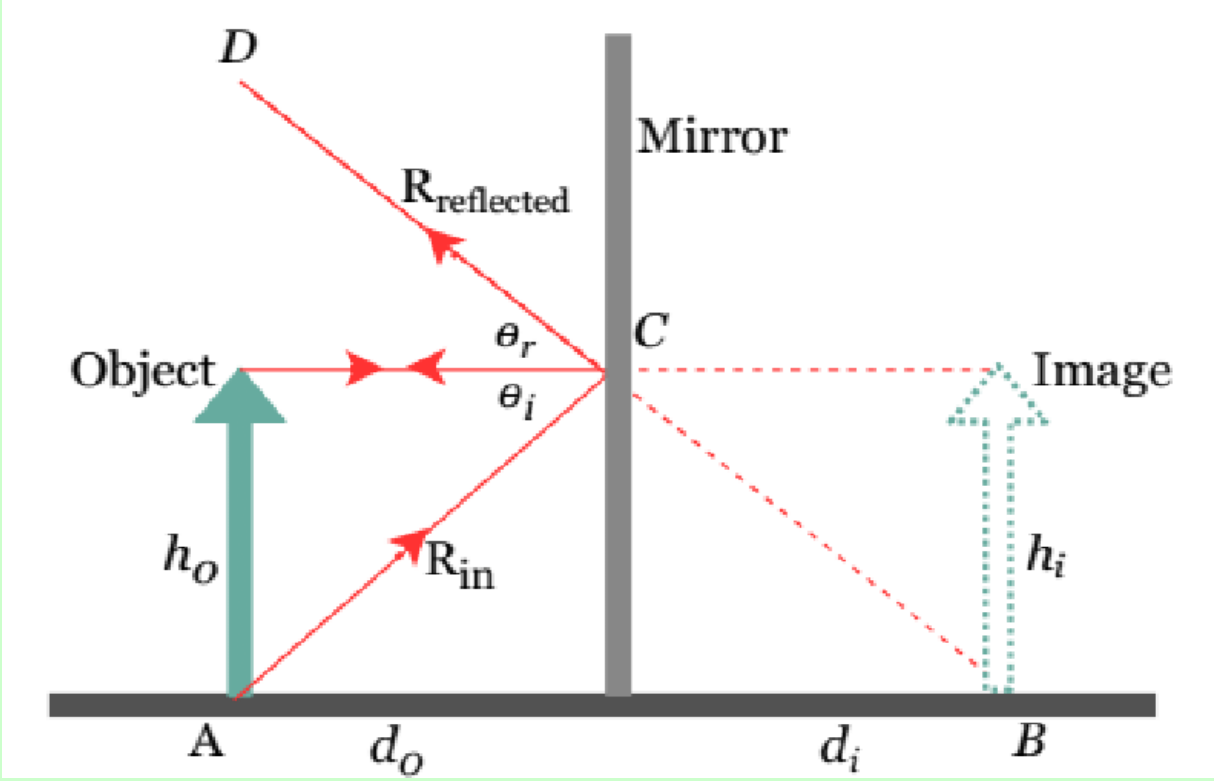
Want a more rigorous geometry…

$d_0 = object\ distance$

$d_i = image\ distance$

$h_0 = object\ height$

$h_i = image\ height$



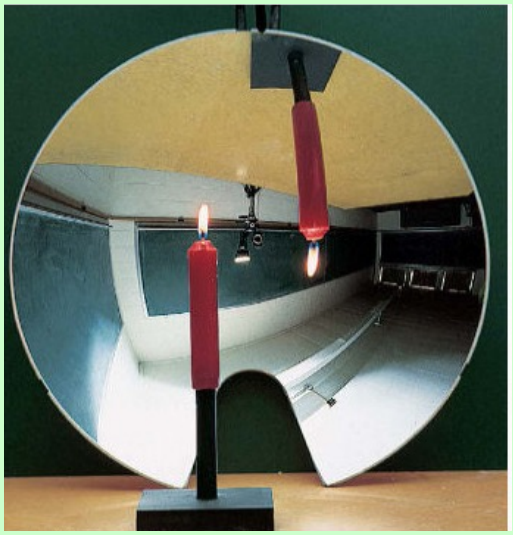
Magnification: $M = \frac{h_i}{h_0} = -\frac{d_i}{d_0}$

How many times bigger/smaller is image compared to object?

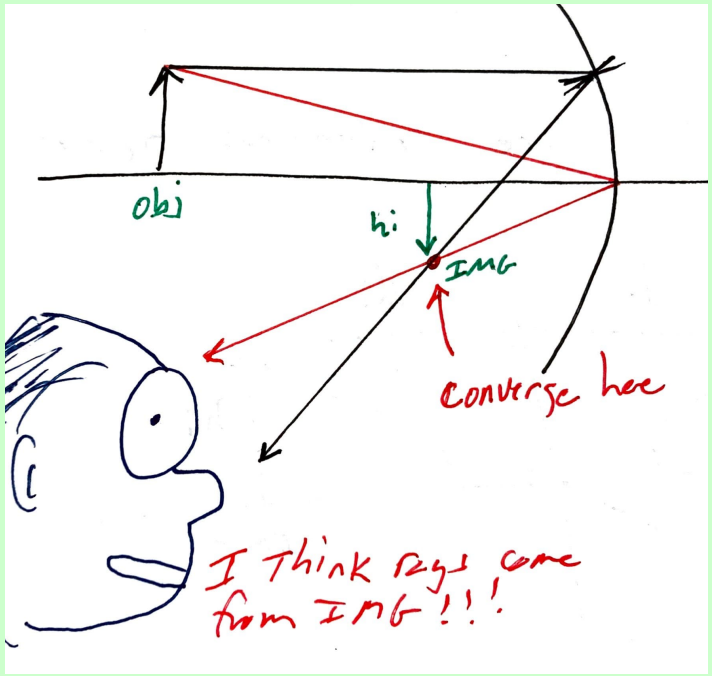
Ex. h_0=5 cm and
. $d_0 = 7$ cm.
Estimate d_i, h_i, M

Spherical Mirrors: because surface is bent, now d_i not equal to d_o

(Think of house of mirrors)



(pick any two rays and follow reflection)



The image is inverted and smaller!

$$M = -\frac{d_i}{d_o} \approx -0.5$$

From geometry you can derive a mirror equation for spherical mirrors

$$\frac{2}{R} = \frac{1}{d_o} + \frac{1}{d_i}$$

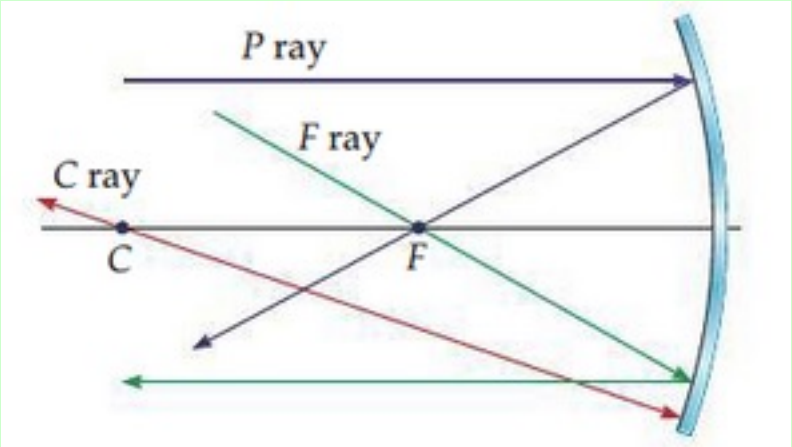
Ex. d_0 = 2 cm. The spherical mirror has a radius of 10 cm. Where is image?

c.f. sec 1.5, pg 2

Many times, (e.g. telescopes, camera) the object is basically at infinity (d_0 = inf)

$$\frac{2}{R} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{2}{R} = \frac{1}{\infty} + \frac{1}{d_i}$$



So all images happen at $d_i = \frac{R}{2}$ This is called the **focal point**, f

The mirror eqn simplifies to:

$$\boxed{\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}}$$

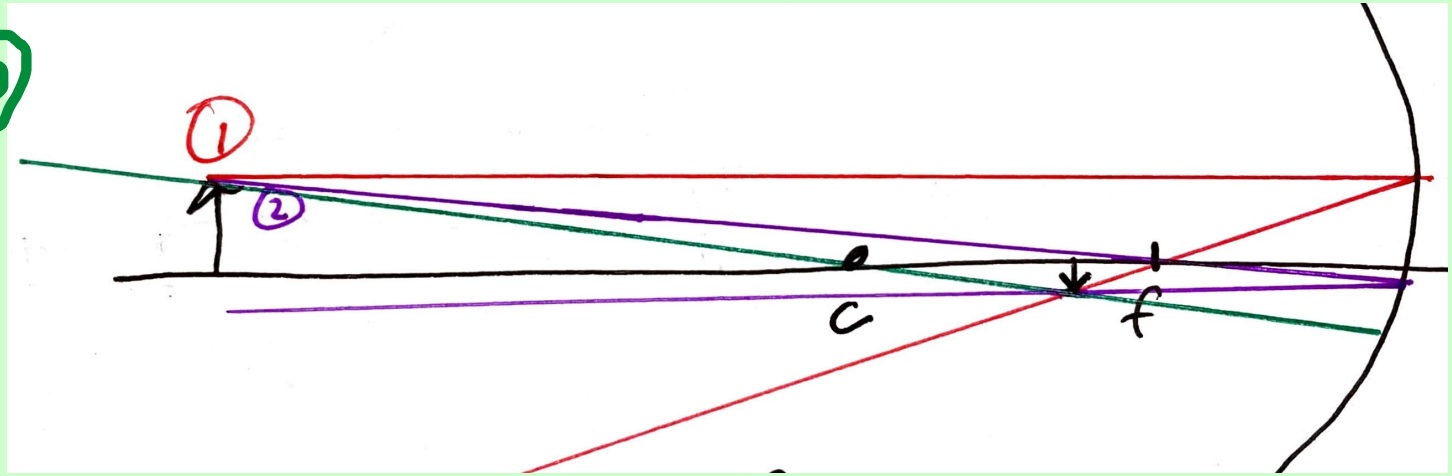
Want to draw clean ray diagrams?

-Draw optical axis, and mirror
-Label points f (=R/2), and
C (center of curv. = R)
-Pick three rays

1. Any parallel rays goes through f
2. Any ray from f goes parallel
3. An ray from C, reflects on self

Quantity	Positive (+)	Negative (-)
d_o	In front of mirror	behind
d_i	In front	Behind
h_o	Upright	Inverted
h_i	Upright	Inverted
m	Upright	Inverted
f	Concave	convex

OA



Ex. do = 8cm, R = 5 cm, ho = 2 cm.

A. Draw ray diagram with three principal rays

B. Estimate di, M, hi. Then calculate precise value.

c.f. Sec 1.5, pg 3

Ex. do = 25cm, Convex, R = 20 cm, ho = 4 cm

c.f. Sec 1.5, pg 4